



Charting the evolution of neuro-symbolic AI in cybersecurity: a scientometric perspective

Shama¹ · Sarika Jain¹

Received: 25 September 2025 / Accepted: 5 February 2026
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026

Abstract

Neuro-symbolic artificial intelligence (NSAI) integrates neural learning with symbolic reasoning to address complex cybersecurity challenges through interpretable and adaptive solutions. This study presents the scientometric mapping of NSAI research in cybersecurity, analyzing publications from Scopus database spanning 2016–2025. Through keyword co-occurrence, bibliographic coupling, and systematic literature review, we identify four thematic clusters and reveal dominant integration paradigms. Our findings show rapid field growth, with Learning-for-Reasoning architectures as the predominant approach. Network intrusion detection and malware analysis emerge as mature domains, while autonomous cyber defense and IoT security remain underexplored. Significant research gaps remain in developing computational efficiency benchmarks, standardized evaluation frameworks, and explainability mechanisms. This integrated approach provides strategic insights for advancing NSAI-driven cybersecurity research and practice.

Keywords Neuro-symbolic AI · Cybersecurity · Scientometric analysis · Bibliometrics · Explainable AI

1 Introduction

Artificial intelligence (AI) has progressed rapidly in recent years, with neural networks and symbolic reasoning systems emerging as two dominant paradigms. While each has achieved significant success, their integration known as neuro-symbolic AI (NSAI) offers a hybrid model that unites the pattern-recognition capabilities of neural networks with the logical inference and interpretability of symbolic AI. Neural-symbolic learning systems harness the complementary strengths of neural networks and symbolic reasoning frameworks [1, 2]. This synergy enables systems to both learn from complex data and reason about patterns, making NSAI particularly suited to high-stakes domains such as cybersecurity. Cybersecurity is a uniquely dynamic and adversarial environment, where threats continually evolve in sophistication and scale. Traditional rule-based defenses tend to be reactive, leaving systems vulnerable to novel attack vec-

tors. NSAI offers the potential for more adaptive, explainable, and resilient security solutions capable of detecting emerging threats, anticipating risks, and providing justifiable reasoning for decisions. Such capabilities are increasingly critical not only for operational resilience but also for meeting regulatory and compliance requirements.

Many key survey papers provide an excellent introduction to this topic. Table 1 summarizes 14 key review papers on Neuro-Symbolic AI. These review papers include the specific application domain of Military applications, Geo-Science, NLP, and Cognitive AI. Despite growing interest, the research landscape of NSAI in cybersecurity remains fragmented, with limited visibility into its intellectual structure, collaborative networks, and thematic evolution. Scientometric and bibliometric methods provide powerful tools for such strategic intelligence, enabling stakeholders to track publication trends, map thematic clusters, identify influential contributors, and detect emerging research frontiers.

This study addresses a significant gap in the literature; to the best of our knowledge, no comprehensive scientometric mapping of neuro-symbolic AI (NSAI) research in cybersecurity has been conducted to date. Drawing on data from Scopus, we combine quantitative mapping techniques, including keyword co-occurrence and bibliographic coupling, with a systematic literature review to produce aca-

✉ Shama
shama_6170013@nitkkr.ac.in

Sarika Jain
jasarika@nitkkr.ac.in

¹ Department of Computer Applications, National Institute of Technology, Kurukshetra, India

demetic and actionable insights. Specifically, we address three guiding research questions:

RQ1: What are the key research themes, application domains, and intellectual structures shaping the field of neuro-symbolic AI in cybersecurity?

RQ2: How has the field evolved over time in terms of publication trends, influential documents and sources, and collaboration structures revealed through bibliographic coupling based analysis?

RQ3: What challenges and future research opportunities emerge from the existing literature on neuro-symbolic AI for cybersecurity?

The rest of this article is structured as follows: Section 2 presents a theoretical overview of neuro-symbolic AI. Section 3 describes the research methodology. Section 4 addresses RQ1 and RQ2 by presenting the scientometric findings, including research themes, application domains, publication trends, and influential authors and networks. Section 5 responds to RQ3 through a systematic literature review, highlighting the key challenges and future opportunities for neuro-symbolic AI in cybersecurity. Finally, Sect. 6 synthesizes the results, offering concluding insights and outlining directions for further research.

2 Theoretical overview

The field of neuro-symbolic AI brings together symbolic logic and neural networks, integrating symbolic and connectionist strategies into a unified framework [2]. To highlight the key differences and complementary strengths of symbolic AI and neural AI, a comparative overview is presented in Table 2. The purpose of this table is to illustrate how each paradigm addresses different aspects of intelligence and to emphasize why their integration forms a powerful hybrid framework in neuro-symbolic AI.

Neural networks are highly effective in identifying intricate patterns within large datasets, yet they often lack mechanisms for explicit knowledge representation and logical inference. In contrast, symbolic reasoning excels in handling structured knowledge and logic-driven tasks but tends to underperform in areas that require learning from data or generalizing across diverse scenarios [7]. Although symbolic methods offer strength in structured logic processing, they can become inefficient when applied to large-scale or complex challenges. Conversely, neural networks thrive in processing vast amounts of data and uncovering hidden patterns. Neuro-symbolic AI seeks to merge the strengths of both paradigms, combining neural learning with symbolic reasoning to develop more resilient, adaptable, and intelligent systems. An example of such a hybrid architecture, illustrated in Fig. 1, demonstrates how integrating symbolic reasoning with neural networks can enhance the system's

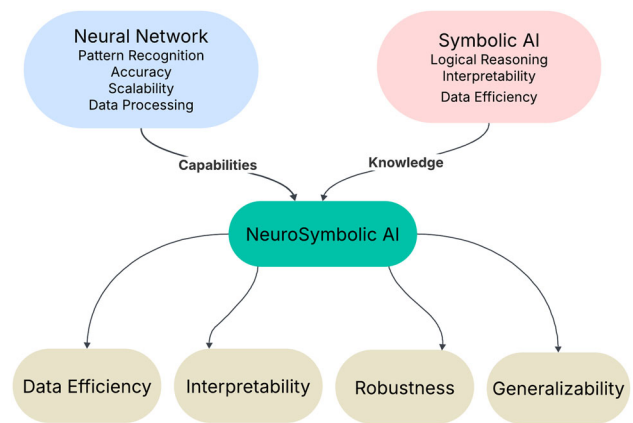


Fig. 1 Neuro-Symbolic AI: hybrid network

decision-making capabilities. From a technology management perspective, this integration addresses a critical trade-off in AI adoption: the need for adaptive, high-performance models that remain interpretable and auditable. In domains like cybersecurity where explainability supports trust, governance, and regulatory compliance, this combination provides an unique value.

Neuro-symbolic AI emerges as a hybrid paradigm that seeks to integrate the structured reasoning of symbolic systems with the adaptive learning capabilities of neural models. In this section, we first explore the foundational components, symbolic AI, neural AI, and their integration by examining their underlying principles, capabilities, and representative tools and frameworks. Following this theoretical foundation, we examine the current cybersecurity problem landscape to understand the challenges and limitations that neuro-symbolic AI aims to address in this domain.

2.1 Symbolic artificial intelligence

Symbolic artificial intelligence focuses on explicit knowledge representation, logical reasoning, and rule-based decision making. Unlike purely data-driven approaches, symbolic methods encode domain expertise in formal structures such as rules, ontologies, and logic programs, enabling transparency, constraint satisfaction, and explainable security analytics [15]. In cybersecurity, symbolic AI is particularly valuable for enforcing policies, reasoning about attack chains, validating decisions, and supporting human-understandable explanations.

The symbolic components reviewed in this section form the conceptual foundation for the “Symbolic Component” columns in the literature analysis tables presented in Sect. 5 *Rule-based and expert systems.*

Rule-based reasoning represents security knowledge using if–then constructs derived from expert insight, standards, and threat intelligence. Such systems enable policy enforcement,

Table 1 Review of reviews in neuro-symbolic AI

References	Year	Focus
De Raedt et al. [3]	2020	Reviews key dimensions of neuro-symbolic systems, including model types, inference methods, logic integration, and representation
Hitzler et al. [4]	2022	Outlines key properties of neuro-symbolic systems, contrasts them with statistical relational AI, and highlights DeepProbLog as an integration example
Wang et al. [5]	2022	Classifies neuro-symbolic systems by integration, representation, embedding, and functionality; benchmarks them on retrosynthesis, semantic parsing, and math problem-solving
Garcez et al. [6]	2023	Provides an overview of neuro-symbolic components, categories, applications, and challenges; emphasizes its role in the Explainable Artificial Intelligence(XAI) paradigm
Sheth et al. [7]	2023	Emphasizes symbolic mappings to improve explainability and control in safety-critical domains like healthcare and autonomous driving
Jalaian et al. [8]	2023	Explores how NSAI enhances IDS by combining learning and reasoning for robust and transparent cybersecurity
Yu et al. [9]	2024	Proposes a taxonomy for neuro-symbolic systems and highlights complementary strengths of neural and symbolic paradigms
Zhang et al. [10]	2024	Overview of Neural-Symbolic AI focusing on representation spaces: multimodal, heterogeneous, and adaptive types
Hagos et al. [11]	2024	Examines ethical, strategic, and technical aspects of deploying NSAI in military contexts
Cheng et al. [12]	2024	Explores NSAI, ML, and DL in geoscience, especially for mineral prediction
Hamilton et al. [13]	2024	Evaluates NSAI in NLP for reasoning, OOD generalization, interpretability, and transfer learning
Wan et al. [14]	2024	Analysis of NSAI models showing symbolic workloads dominate runtime and scalability issues
Bhuyan et al. [15]	2024	Examines representation, learning, reasoning, and decision making in NSAI with applications
Hakim et al. [16]	2025	Systematic review of NSAI in cybersecurity focusing on approaches, frameworks, and challenges

alert correlation, and explainable decision logic. In NSAI cybersecurity, rules are commonly used to validate neural predictions, guide action selection, and encode mitigation strategies observed in intrusion detection and autonomous defense platforms.

Ontologies and knowledge graphs.

Ontologies define structured vocabularies and semantic relationships between cyber assets, vulnerabilities, threats, and countermeasures. Implemented using languages such as Web Ontology Language(OWL) and Semantic Web Rule Language(SWRL), ontologies enable semantic interoperability and logical inference. Knowledge graphs extend ontologies by integrating large-scale, heterogeneous cybersecurity data, supporting reasoning over attack paths, vulner-

ability dependencies, and contextualized threat intelligence, as reflected in several NSAI systems surveyed in Sect. 5.

First-order logic and answer-set programming.

First-order logic (FOL) and answer-set programming (ASP) provide formal reasoning frameworks for modeling constraints, policies, and attack semantics. FOL supports quantification and relational reasoning, while ASP enables non-monotonic reasoning under uncertainty. These mechanisms are frequently coupled with neural models to enforce logical consistency, explain predictions, and support planning in autonomous cyber defense.

Fuzzy logic and probabilistic logic.

Cyber environments are inherently uncertain and noisy. Fuzzy logic extends classical logic with degrees of truth, enabling approximate reasoning in anomaly detection and

Table 2 Comparison of characteristics of neural and symbolic systems

Characteristic	Neural networks	Symbolic AI
Learning Style	Data-driven learning from examples	Rule-based, knowledge-driven reasoning
Knowledge Representation	Distributed, implicit in weights	Explicit, formal symbolic structures
Interpretability	Low (black-box behavior)	High (transparent reasoning process)
Data Requirements	High (needs large labeled datasets)	Low (can operate with predefined knowledge)
Adaptability	High adaptability to new data	Limited adaptability without manual updates
Handling Noisy Data	Generally robust	Sensitive to noise and ambiguous inputs
Scalability	High (with enough compute resources)	Limited scalability (manual knowledge engineering)
Typical Use Cases	- Image classification - Speech recognition - Expert systems	- Planning - Language modeling - Theorem proving
Strengths	- Pattern recognition - Scalability - Interpretability	- Logical reasoning - Robustness to noise - Data efficiency
Main Limitations	- Poor logical reasoning - Lack of transparency - Knowledge acquisition bottleneck	- Brittle to real-world variability - Requires massive data - Poor pattern recognition

risk assessment. Probabilistic logic programming (PLP) further incorporates uncertainty into symbolic reasoning, supporting vulnerability prioritization and decision making under incomplete information [17].

Formal verification and constraint reasoning.

Formal verification techniques ensure that learned security policies satisfy safety, compliance, and operational constraints. By integrating symbolic constraints with neural learning, NSAI systems can verify correctness, prevent unsafe actions, and maintain policy compliance. Such mechanisms are particularly relevant for safety-critical domains such as industrial control systems and autonomous cyber defense agents reviewed later in the manuscript.

These symbolic paradigms supply the reasoning backbone for NSAI cybersecurity architectures. Their integration with neural learning components, as systematically reviewed in Sect. 5, enables explainable, constrained, and policy-aware intelligent security systems.

2.2 Neural architectures

Neural artificial intelligence emphasizes data-driven learning, representation extraction, and adaptive pattern recognition. In cybersecurity, neural models learn from network traffic, logs, binaries, and behavioral traces to identify malicious activity, predict vulnerabilities, and optimize defense strategies. These neural architectures constitute the “Neural Component” dimension in the literature review tables presented in Sect. 5, where they are systematically combined with symbolic reasoning modules to form NSAI frameworks.

Deep neural networks and convolutional models.

Deep neural networks (DNNs) and convolutional neural networks (CNNs) are widely used for feature learning from raw cybersecurity data such as packets, binaries, and system calls. CNNs capture spatial and structural patterns, enabling malware image analysis, traffic classification, and behavioral fingerprinting in intrusion detection systems.

Recurrent and sequential models.

Recurrent neural networks (RNNs), including LSTM and GRU architectures, model temporal dependencies in sequential security data. They are particularly effective for anomaly detection, user behavior modeling, and attack sequence prediction, forming a core component of several NSAI intrusion detection and threat intelligence frameworks surveyed in Sect. 5.

Graph neural networks and representation learning.

Graph neural networks (GNNs), temporal GNNs (TGNNs), and variational graph autoencoders (VGAE) operate on graph-structured cybersecurity data such as communication graphs, call graphs, and dependency networks. These models support relational learning, attack path modeling, and knowledge-infused representation learning in modern NSAI platforms.

Autoencoders and generative models.

Autoencoders learn compact representations for anomaly detection, while generative adversarial networks (GANs) support synthetic data generation and adversarial training. Knowledge-guided variants, as reviewed later, enable NSAI systems to generate realistic yet policy-consistent security data for training and robustness enhancement.

Reinforcement learning and deep RL.

Reinforcement learning (RL) and deep reinforcement learning (DRL) optimize sequential decision making in dynamic cyber environments. Algorithms such as deep Q-Network (DQN) and proximal policy optimization (PPO) learn defense policies through interaction, enabling autonomous response, attack-defense simulation, and resource allocation when integrated with symbolic planning and constraints.

Federated learning and foundation models.

Federated learning supports decentralized training across distributed security agents without exposing raw data, addressing privacy and scalability challenges. In parallel, foundation models and large language models (LLMs) provide semantic abstraction, code understanding, and contextual reasoning capabilities, increasingly serving as cognitive layers within NSAI cybersecurity systems.

Collectively, these neural paradigms supply the perceptual and adaptive capabilities of NSAI. Their principled integration with symbolic reasoning components, as analyzed in Sect. 4, underpins the construction of scalable, explainable, and intelligent cyber defense architectures.

2.3 Neuro-symbolic AI: best of both worlds

Contemporary research reflects a renewed focus on neuro-symbolic paradigms, which seek to integrate the pattern recognition capabilities of neural networks with the logical reasoning strengths of symbolic systems, thereby addressing limitations inherent in each approach when used in isolation. Henry Kautz [18] proposed a taxonomy distinguishing integration strategies based on coupling strength, while authors in [7] emphasized practical motivations to combine these paradigms in real-world applications. Besold *et al.* [19] provided a comprehensive survey, categorizing historical and modern integration techniques. Further, Garcez *et al.* [6] developed formal frameworks for neural-symbolic reasoning, and Marcus *et al.* [20] advocated for hybrid intelligence to address limitations in purely data-driven models. Based on these foundational efforts, Yu *et al.* [9] recently proposed a comprehensive taxonomy that systematically classifies neural-symbolic systems according to integration levels, learning strategies, and reasoning capabilities, as discussed in the following section.

2.3.1 Learning for reasoning (L4R)

The first category, commonly termed learning for reasoning, aims to combine the strengths of neural networks with symbolic reasoning systems to enhance problem-solving capabilities. In this approach, the model follows a sequential structure where the neural component operates first, extracting meaningful features or symbolic representations from the input data. These outputs are then passed to the symbolic rea-

soning module, which performs more advanced, logic-based reasoning tasks.

2.3.2 Reasoning for learning (R4L)

The second category, known as reasoning for learning, involves the use of symbolic systems to enhance and guide the learning process of neural networks. In this framework, symbolic knowledge is integrated into the neural architecture, with the neural network serving as the primary decision-making component. Unlike sequential models, this approach features a parallel structure in which both symbolic and neural components function simultaneously. While the neural network learns patterns from the data, the symbolic system contributes structured knowledge or constraints that help shape and refine the learning process.

2.3.3 Learning–reasoning (L–R)

The third category, called learning–reasoning, features a bidirectional interaction between neural and symbolic systems, where both components collaborate equally and contribute jointly to the problem-solving process. In this approach, neural networks and symbolic reasoning are integrated in an alternating manner, allowing them to exchange information and influence each other iteratively. This cooperation enables the model to leverage the learning capabilities of neural systems alongside the structured logic of symbolic reasoning to produce a unified solution. These hybrid approaches offer both adaptability and interpretability qualities for cybersecurity, a domain characterized by high uncertainty, rapid adversary adaptation, and stringent compliance requirements.

2.4 Cybersecurity: problem landscape

Cybersecurity is a multidisciplinary field dedicated to protecting digital assets, systems, and data from malicious activities. The proliferation of cloud platforms, IoT devices, and critical infrastructures has elevated cybersecurity from a technical discipline to a strategic imperative for governments, industries, and society. Drawing on computer science, behavioral science, and formal methods, cybersecurity aims to ensure confidentiality, integrity, and availability of systems, while maintaining accountability, resilience, and regulatory compliance.

Traditional rule-based and statistical methods are increasingly challenged by the complexity of modern threats, prompting adoption of AI and ML for adaptive, data-driven protection. However, the opaqueness of purely neural models poses trust, interpretability, and safety concerns, making NSAI a promising alternative [1]. Recent studies [21] highlight how lightweight and explainable AI solutions are

increasingly required in resource-constrained real-time environments such as edge networks, settings where NSAI's hybrid reasoning capabilities are particularly advantageous.

Table 3 presents a comprehensive taxonomy of cybersecurity subdomains organized into six strategic clusters. Each cluster addresses distinct security challenges where NSAI integration offers unique advantages through the combination of adaptive learning and interpretable reasoning. This structured mapping supports decision-makers in identifying where NSAI capabilities can create the greatest operational and strategic value, guiding both internal development and external partnerships. This taxonomy serves as the conceptual framework for analyzing NSAI implementations across different application domains in subsequent sections.

This taxonomic organization provides a structured framework for systematically analyzing how NSAI approaches address different cybersecurity challenges. The classification enables the assessment of where hybrid neural-symbolic architectures deliver the greatest operational and strategic value, guiding both research priorities and practical deployment decisions in the field.

3 Research methodology

This section describes the methodological framework adopted to investigate neuro-symbolic AI (NSAI) research in cybersecurity. It outlines the overall research design, data collection and screening process, scientometric analysis procedures, and systematic literature review (SLR) protocol used to ensure comprehensive, transparent, and reproducible analysis of the literature.

3.1 Study design

Neuro-symbolic AI in cybersecurity constitutes an emerging interdisciplinary domain characterized by rapid technical evolution, heterogeneous publication venues, and inconsistent terminological conventions. These characteristics necessitate a methodological approach that captures both the breadth of the research landscape and the depth of technical contributions. Accordingly, this study adopts a dual-methodology framework integrating scientometric analysis with a systematic literature review (SLR).

Scientometric analysis employs quantitative bibliometric techniques to map the intellectual structure of the field through examination of publication metadata, citation relations, and collaboration networks. It encompasses performance analysis, including publication trends, citation impact, and institutional contributions, and science mapping, which reveals thematic clusters, co-authorship patterns, and knowledge evolution through keyword co-occurrence and co-citation analysis. Applied to the complete corpus of identified

studies ($n = 262$), this phase provides macro-level insights into research trajectories, dominant themes, and collaborative dynamics.

Complementing this, the SLR follows rigorous protocols for identifying, evaluating, and synthesizing technical research. Conducted on a curated high-quality subset ($n = 63$), the SLR enables micro-level examination of neuro-symbolic architectures, integration strategies, implementation methodologies, and empirical outcomes in cybersecurity. This qualitative synthesis explains not only what approaches are adopted, but also how and why specific design choices prove effective in security contexts.

Integrating these complementary methodologies addresses limitations inherent in single-method studies. While existing surveys offer technical synthesis, they often lack quantitative insight into field structure and temporal evolution. Conversely, purely bibliometric studies may reveal trends without clarifying underlying technical mechanisms. The proposed framework unifies breadth and depth by combining quantitative mapping with qualitative technical analysis, which is particularly valuable for fragmented and rapidly evolving domains such as NSAI in cybersecurity.

The methodology follows established standards for bibliometric research and adheres to PRISMA 2020 guidelines for systematic reviews, ensuring transparency, reproducibility, and methodological rigor throughout the research process.

3.2 Data source selection

This study utilized the Scopus database for comprehensive scientometric analysis. Scopus, operated by Elsevier, is one of the largest abstract and citation databases, encompassing over 23,000 peer-reviewed journals, conference proceedings, and book series across scientific, technical, medical, and social sciences disciplines. Scopus was selected due to its extensive coverage of science literature, standardized metadata for scientometric analysis, and robust analytical tools including author profiles, institutional metrics, h-index calculations, and citation tracking. The database's widespread use in scientometric studies ensures comparability with the existing research in this domain.

3.3 Query development and validation

The search query in Table 6 was developed through a three-stage iterative process to balance recall and precision.

Stage 1: Keyword identification

Keywords were derived from foundational NSAI surveys (Table 1) and domain expert consultation and grouped into two semantic categories as shown in Table 4.

This taxonomy ensured comprehensive coverage across both methodological approaches (neuro-symbolic AI) and application domains (cybersecurity).

Table 3 Cybersecurity subdomain taxonomy

Subdomain	Key focus areas	NSAI relevance
2.4.1 Cluster - Defensive Operations & Threat Management		
Network Security	Protocol protection, traffic analysis, DoS/DDoS defense	First line of defense; neural pattern detection + symbolic policy enforcement
Threat Detection & Response	Malware detection, anomaly detection, insider threats, behavioral profiling	Real-time detection with explainable reasoning chains
Threat Prediction & Proactive Defense	Cyber attack prediction, anomaly forecasting, threat intelligence modeling	Combines pattern recognition with logical reasoning for anticipatory defense
Digital Forensics & Incident Response	Post-incident evidence gathering, forensic analysis, structured response	Transparent investigation trails supporting compliance
2.4.2 Cluster - Vulnerability Assessment & Risk Management		
Vulnerability & Risk Management	Vulnerability scanning, static/dynamic code analysis, penetration testing	Symbolic constraints encode policies; neural components identify novel patterns
Security Governance, Risk & Compliance (GRC)	GDPR, HIPAA, ISO/IEC standards alignment	Auditable decision trails through symbolic reasoning
2.4.3 Cluster - Domain-Specific Security Applications		
Web, Cloud & API Security	API protection, SQL injection, XSS, insecure configurations	Adaptive reasoning for dynamic, distributed environments
Cyber-Physical Systems (CPS) Security	ICS, medical devices, smart grids	Interpretable and verifiable AI for safety-critical contexts
Identity, Access Control & Authentication	Access control models, MFA, zero-trust architectures	Policy encoding with adaptive threat response
Data Security & Privacy	Encryption, privacy-preserving computation, differential privacy	Complex policy representation with neural adaptation
2.4.4 Cluster - AI-Centric Security Challenges		
Explainability, Reasoning & AI Safety	Transparent AI systems, symbolic rules, logic-based methods	Bridges neural learning with symbolic explainability
Adversarial ML & Model Security	Adversarial examples, data poisoning, model inversion	Symbolic constraints encode invariants
2.4.5 Cluster - Operational Automation & Human Factors		
Security Automation & Orchestration	SIEM, SOAR, automated response	Adaptive learning with explainable decisions
Social & Human-Centric Security	Phishing, social engineering, behavioral monitoring	Human-in-the-loop security
2.4.6 Cluster - Emerging Research Frontiers		
Emerging Areas	Blockchain security, quantum-safe crypto, cyber-deception	Adaptive learning + explainable decisions

Table 4 Keyword categories for query construction

Category	Terms and variants
Neuro-Symbolic AI	neuro-symbolic, neural-symbolic, neurosymbolic neuro symbolic, neural symbolic (hyphenation variants) hybrid AI, symbolic-connectionist, symbolic connectionist deep symbolic learning (“neural networks” AND “symbolic”) [conjunctive phrase]
Cybersecurity	cybersecurity, cyber security, information security digital security, network defense, cyber defense intrusion detection, intrusion prevention cyber threat, threat intelligence, threat detection threat mitigation, cyber attack malware, vulnerability, privacy and security, cyber agent

Table 5 Query variant pilot testing results

Variant	Query structure	Records	Assessment
A (Narrow)	TITLE-ABS-KEY(("neuro-symbolic" OR "neural-symbolic") AND ("cybersecurity" OR "cyber security" OR "cyber-security"))	33	Insufficient coverage; missed relevant hybrid AI studies
B (Moderate) SELECTED	Complete query as shown in Table 6 with all NSAI and cybersecurity terms	262	Optimal balance between precision and recall
C (Broad)	[Query B] OR ("explainable AI" OR "interpretable AI")	2400	Excessive noise; included general XAI papers without neuro-symbolic focus

Bold formatting is used to indicate the selected/recommended query variant

Stage 2: Pilot testing and query optimization

To identify the optimal query formulation, we conducted pilot tests using three query variants with different levels of specificity. The pilot searches were executed on December 31, 2025, spanning publications from January 1, 2016, to December 31, 2025, to establish baseline retrieval statistics and assess query performance.

Stage 3: Validation and precision assessment

To validate the selected Query Variant B, we performed manual quality assessment through stratified sampling. A random sample of 30 records retrieved by Query B was independently reviewed by both the authors against the following relevance criteria:

1. Explicit mention of neuro-symbolic integration, hybrid neural-symbolic architectures, or equivalent concepts
2. Application to cybersecurity domain (threat detection, vulnerability analysis, intrusion detection, etc.)
3. Peer-reviewed status (journal articles, conference papers, or workshop proceedings)

Outcome: 26 out of 30 records (86%) met all three criteria, indicating high precision. The three excluded records were

tangentially related works on general symbolic AI without neural integration (Table 5).

Final query formulation and reproducibility

Based on the validation results, Query Variant B was selected for the complete systematic search. The final query string, as shown in Table 6, balances comprehensive coverage with acceptable precision. To ensure reproducibility (Table 7):

- **Search Date:** 31 December 2025
- **Database:** Scopus (Elsevier)
- **Search Fields:** TITLE-ABS-KEY (Title, Abstract, Keywords)
- **Temporal Coverage:** 2016–2025
- **Language:** Only English
- **Document Types:** No restrictions at query stage (filtered during screening)

3.4 Study selection and PRISMA screening

A four-phase PRISMA 2020 screening protocol [22] was applied to systematically filter the retrieved records and construct the final review corpus. Two independent reviewers (Author 1: PhD candidate; Author 2: Faculty member with

Table 6 Scopus search query components for neuro-symbolic AI in cybersecurity

Query component	Search terms
Neuro-Symbolic Concepts	"neuro-symbolic" OR "neurosymbolic" OR "neural-symbolic" OR "neuro symbolic" OR "neural symbolic" OR "hybrid AI" OR "hybrid intelligence" OR "knowledge-guided learning" OR "knowledge-guided neural" OR "knowledge guided" OR "knowledge graph neural" OR "knowledge graph deep learning" OR "symbolic reasoning neural" OR "symbolic connectionist" OR "deep symbolic learning"
Boolean Operator	AND
Cybersecurity Concepts	"cybersecurity" OR "cyber security" OR "cyber-security" OR "information security" OR "network security" OR "computer security" OR "network defense" OR "cyber defense" OR "cyber threat" OR "intrusion detection" OR "intrusion prevention" OR "threat detection" OR "threat intelligence" OR "threat mitigation" OR "threat analysis" OR "malware" OR "malware detection" OR "malware analysis" OR "vulnerability" OR "vulnerability detection" OR "vulnerability analysis" OR "cyber attack" OR "cyber-attack" OR "cyber-attack" OR "attack detection" OR "security automation" OR "security analytics" OR "privacy and security" OR "cyber agent" OR "digital security"
Search Fields	TITLE-ABS-KEY (Title, Abstract, Keywords)

expertise in AI) conducted screening at both the title/abstract and full-text levels to ensure consistency and methodological rigor. Figure 2 illustrates the complete selection workflow from initial database query to final corpus construction.

Phase 1 (Identification). The initial search retrieved 262 records from the Scopus database using the validated query described in Sect. 3.3. As only a single database was used, no duplicate removal was required, and all 262 records were advanced to the screening stage.

Phase 2 (Title and Abstract Screening). In this phase, records were evaluated for topical relevance to neuro-symbolic AI in cybersecurity. Studies were excluded if they (i) relied purely on conventional ML/DL without symbolic integration, (ii) belonged to non-cybersecurity domains, (iii) represented inappropriate document types (e.g., proceedings listings without technical content), or (iv) were general surveys lacking methodological contribution. This step resulted in the exclusion of 134 records and the advancement of 128 studies to full-text assessment.

Phase 3 (Full-Text Eligibility Assessment). The remaining studies were examined in detail to verify genuine neuro-symbolic integration and sufficient technical depth. Exclusions were applied to works labeled as “hybrid” without symbolic reasoning, studies adopting alternative AI paradigms (e.g., physics-informed learning), and papers with insufficient implementation or evaluation detail for comparative analysis. This stage excluded 65 records, yielding a final corpus of 63 eligible studies.

Phase 4 (Final Corpus Construction). The resulting 63 studies formed the basis for both scientometric analysis and systematic literature review. This corpus was used for quantitative bibliometric mapping (Sect. 4) and qualitative thematic synthesis of neuro-symbolic architectures and cybersecurity applications (Sect. 5).

4 Scientometric analysis

Scientometrics employs quantitative methods to examine patterns within scientific literature, enabling systematic analysis of citation dynamics, research impact assessment, and knowledge evolution within specific domains [23]. Unlike traditional narrative reviews that may inadvertently overlook conceptual relationships or introduce subjective bias, scientometric approaches leverage large-scale bibliometric datasets to reveal systematic patterns, collaboration networks, and intellectual structures that emerge from collective scholarly activity [24]. This scientometric analysis is designed to systematically address the research questions guiding this study. In particular, RQ1 investigates the key research themes, application domains, and intellectual structures shaping neuro-symbolic AI in cybersecurity. RQ2 examines how the field has evolved over time, including publication trends, influential documents, and collaboration structures revealed through bibliographic coupling analysis. Finally, insights derived from the scientometric findings inform RQ3 by identifying structural gaps, emerging challenges, and future research opportunities discussed in the subsequent systematic review section.

4.1 Understanding VOSviewer visualization and metrics

VOSviewer (version 1.6.20) is a software tool for constructing and visualizing bibliometric networks, built upon distance-based mapping techniques that position related items closer together in two-dimensional space [25]. The tool employs the VOS (Visualization of Similarities) mapping

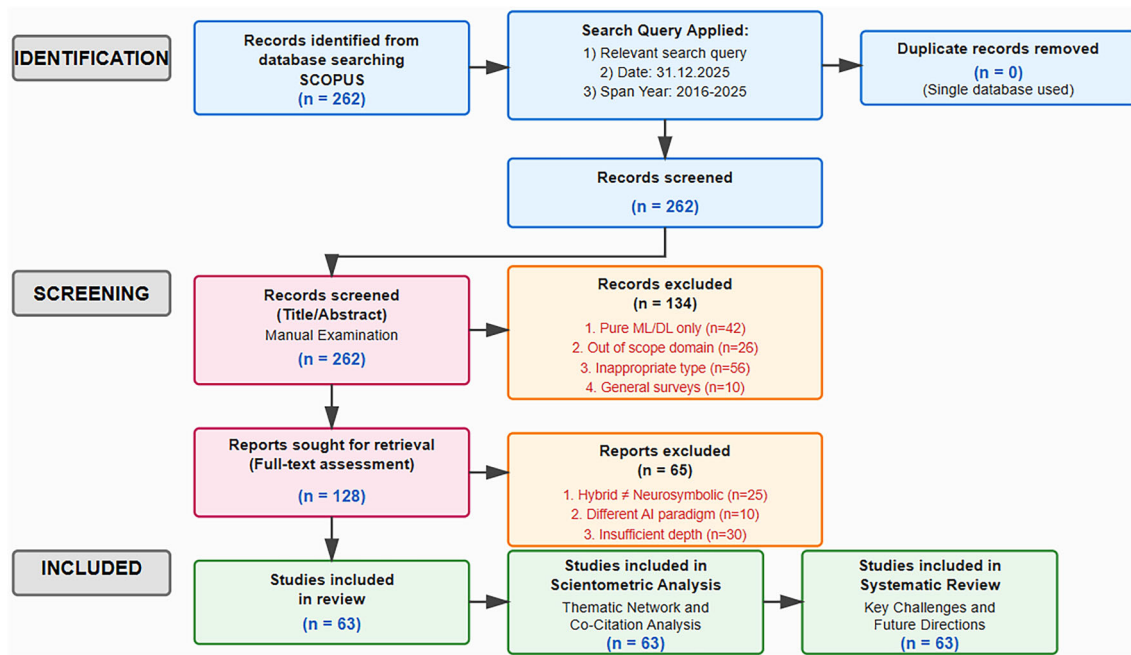


Fig. 2 Systematic selection process

method, which optimizes node placement such that the distance between items reflects their similarity based on co-occurrence strength. In keyword co-occurrence networks, node entities represent author keywords that are scientifically analyzed to reveal thematic structure, knowledge integration patterns, and research evolution within the domain.

Node size in the visualization varies proportionally to occurrence frequency, with larger nodes representing keywords appearing more frequently across the literature, thereby indicating foundational or extensively researched concepts. Node color denotes cluster membership, determined through modularity-based community detection using the smart local moving algorithm [26]. This clustering method partitions the network into thematic groups by optimizing modularity, which reflects how strongly nodes are connected within the same cluster compared with their connections to nodes in other clusters.

Edge thickness signifies co-occurrence strength between keyword pairs, with thicker edges indicating keywords that frequently appear together in publications, thus revealing conceptual associations and interdependencies. Nodes with high degree centrality represent hub keywords that co-occur with numerous other concepts, playing integrative roles in connecting diverse research themes. Such central nodes often bridge multiple clusters, signifying conceptual crossroads where different methodological approaches, application domains, or theoretical frameworks converge.

According to the research objectives and analysis requirements, the utilized general parameter settings of the Vosvie tool are described in Table 8.

The selection of minimum occurrence thresholds (e.g., minimum 4 occurrences for keywords) ensures statistical significance while retaining emerging themes that may represent nascent research directions. The clustering resolution parameter (typically set at 1.0) was selected to provide optimal granularity to identify distinct research themes without excessive fragmentation. Association strength normalization was employed to account for frequency differences across items, providing more accurate similarity measures compared to raw co-occurrence counts.

4.2 Temporal evolution of literature growth over time

Figure 3 illustrates the temporal trajectory of NSAI publications in cybersecurity. The temporal distribution analysis reveals a distinct evolutionary pattern in neuro-symbolic AI cybersecurity research spanning 2016–2025. The field demonstrates three identifiable phases of development.

• Phase 1: Emergence (2016–2020)

The initial period represents a nascent stage with minimal research activity. Annual publications remained consistently low, ranging from 1–3 papers per year. The cumulative publication count grew slowly from 2 to approximately 6 publications by 2020, indicating limited early adoption and exploratory work. This period reflects the foundational stage where researchers began recognizing the potential synergy between neuro-symbolic approaches and cybersecurity applications.

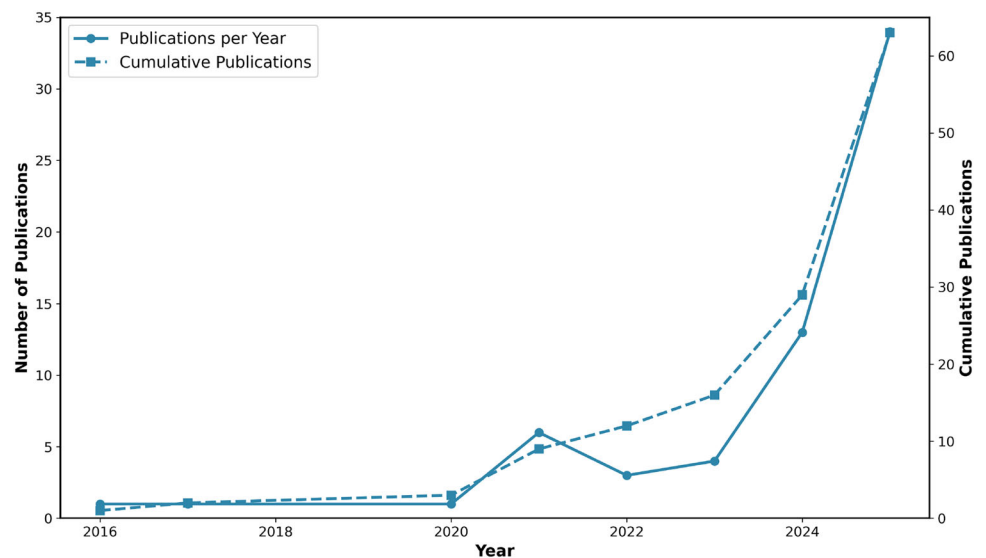
Table 7 PRISMA flow summary with representative exclusion examples

Phase	Description	<i>n</i>	Representative example (exclusions)
Phase 1: Identification			
Records identified	Scopus retrieval	262	—
Duplicate records	None (single database)	0	—
Records to screen	Advanced to screening	262	—
Phase 2: Title/Abstract Screening (n=262)			
<i>Exclusion Reasons:</i>			
Reason 1	Pure ML/DL only	42	“AI-Powered Malware Analysis in Military Cybersecurity: A Deep Learning Approach”
Reason 2	Out-of-scope domain	26	“On the added value of sequential deep learning for upscaling evapotranspiration”
Reason 3	Inappropriate type	56	“26th International Conference on Principles and Practice of Multi-Agent Systems”
Reason 4	General surveys	10	“Modern Approaches to Software Vulnerability Detection: A Survey of ML, DL, and LLMs”
	Total excluded	134	
	Advanced to full-text	128	
Phase 3: Full-Text Assessment (n=128)			
<i>Exclusion Reasons:</i>			
Reason 1	Hybrid $\neq$ neurosymbolic	25	“AI-Driven Fraud Detection: Hybrid Edge-Cloud Framework” / “AI-powered malware detection: hybrid CNN-RNN model”
Reason 2	Different AI paradigm	10	“Physics-informed machine learning in cyber-attack detection and resilient control”
Reason 3	Insufficient depth	30	“A Systematic Review on Hybrid AI Models Integrating Machine Learning and Federated Learning”
	Total excluded	65	
	Met inclusion criteria	63	
Phase 4: Final Corpus			
Scientometric corpus	Network analysis	63	Quantitative bibliometric mapping
Systematic review	Content synthesis	63	Qualitative thematic analysis

Table 8 VOSviewer parameter settings and configuration

Parameter	Setting	Justification
Software	VOSviewer 1.6.20	Latest stable version for bibliometric analysis
Analysis Type	Co-occurrence analysis	To identify thematic relationships among keywords
Unit of Analysis	Author keywords	More specific than index keywords, reflects author intent
Counting Method	Full counting	Each co-occurrence counted equally regardless of document
Minimum Occurrences	4	Balance between comprehensiveness (retaining emerging themes) and clarity (removing noise)
Items Selected	50 of 750 keywords	Represents core research vocabulary with sufficient frequency
Clustering Resolution	1.0	Default resolution provided optimal granularity for domain structure
Normalization	Association strength	Accounts for frequency differences, more accurate than raw co-occurrence
Clusters Identified	4	Natural thematic divisions emerged from algorithm
Modularity (Q)	0.1701	Emerging interdisciplinary domains

Fig. 3 Temporal distribution of records in Neuro-Symbolic AI in Cybersecurity



- **Phase 2: Early Growth (2021–2023)**

A moderate acceleration occurred during this transitional period. The cumulative total reached approximately 19 publications by the end of 2023. This phase suggests growing interest as researchers established initial frameworks and demonstrated proof-of-concept applications, validating the approach’s viability for cybersecurity challenges.

- **Phase 3: Exponential Expansion (2024–2025)**

The most dramatic growth characterizes the recent period. Publications surged from approximately 13 papers in 2024 to 34 papers in 2025 which represents a 162% year-over-year increase. The cumulative total reached 63 publications by 2025, with the trajectory indicating sustained momentum. This exponential growth reflects the field’s maturation, increased funding, broader researcher engagement, and demonstrated practical applications in addressing complex cybersecurity threats requiring both learning and reasoning capabilities.

4.3 Thematic network

Thematic network analysis was applied to uncover the conceptual and application-oriented structure of neuro-symbolic AI research in cybersecurity, directly addressing RQ1 by identifying dominant research themes, integration paradigms, and operational domains. Figure 4 presents the bibliometric co-occurrence network, illustrating how key terms cluster and interact across artificial intelligence, cybersecurity, and vulnerability analysis.

The network consists of 50 nodes representing distinct keywords, where node size reflects term frequency and edge thickness indicates co-occurrence strength. Color-coded communities capture thematic groupings, while the over-

all structure exhibits substantial inter-cluster connectivity (modularity $Q = 0.170$), suggesting an integrative research landscape rather than isolated subfields. This topology indicates that effective neuro-symbolic cybersecurity solutions rely on cross-domain synthesis of learning paradigms, security applications, and symbolic reasoning.

Table 9 reports the top-15 keywords ranked by occurrence and degree centrality. Core security concepts such as “network security” (28), “cyber security” (25), and “cybersecurity” (19) dominate the network, confirming the application-driven orientation of the field. Methodologically significant terms including “deep learning,” “intrusion detection,” “symbolic reasoning,” and “neuro-symbolic AI” further highlight the fusion of adaptive learning with structured reasoning. High centrality values for these terms demonstrate their bridging role across thematic domains, reinforcing the integrative nature of NSAI-based cybersecurity research.

4.4 Keyword cluster analysis

Community detection on the keyword co-occurrence network identifies four thematic clusters with modest modularity ($Q = 0.170$), reflecting strong interaction between research themes. Such connectivity is typical of emerging interdisciplinary fields where progress depends on systematic integration rather than thematic isolation. The main characteristics of each cluster are summarized below.

Cluster 1: Advanced Learning Paradigms and Cyber Defense (n=14) emphasizes neural and hybrid learning methods, including deep and reinforcement learning, federated learning, knowledge graphs, and language models, applied to malware analysis and cyber defense. This cluster represents the adaptive learning component of neuro-symbolic architectures.

Fig. 4 Keyword co-occurrence network revealing four thematic clusters: Cluster 1 (red)—Advanced Learning Paradigms and Cyber Defense; Cluster 2 (green)—Network Security and Threat Detection; Cluster 3 (blue)—AI Foundations and Interpretability; Cluster 4 (yellow)—Complex Systems and Network Infrastructure. Node size reflects occurrence frequency, link thickness indicates co-occurrence strength, and colors denote cluster membership (modularity $Q = 0.170$). (color figure online)

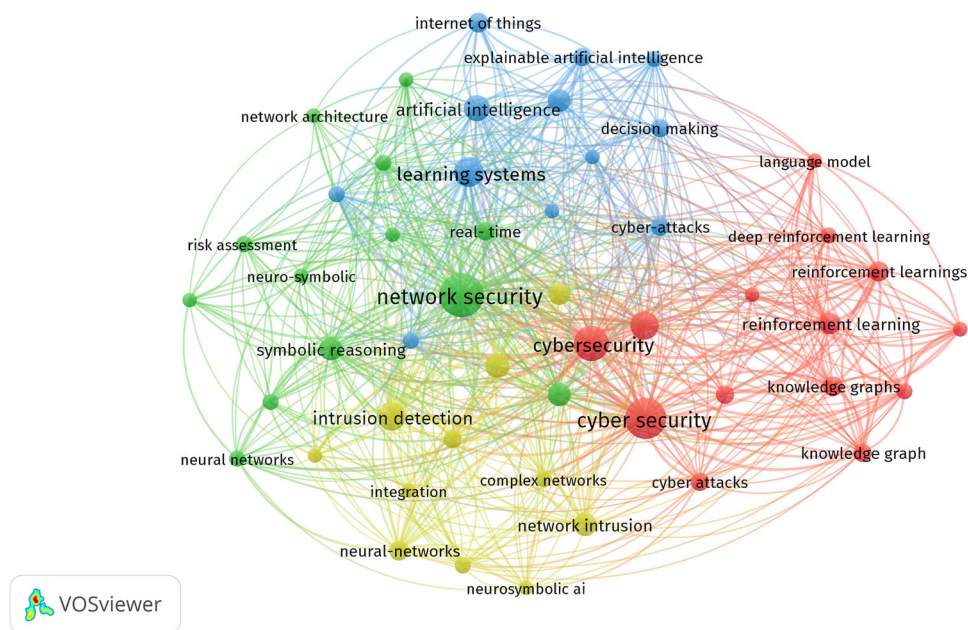


Table 9 Top-15 keywords by co-occurrence frequency and degree centrality

Keyword	Occurrences	Keyword	Centrality
Network security	28	Network security	0.98
Cyber security	25	Cyber security	0.92
Cybersecurity	19	Cybersecurity	0.92
Learning systems	14	Deep learning	0.82
Deep learning	13	Learning systems	0.82
Intrusion detection	13	Cyber threats	0.78
Artificial intelligence	11	Computer crime	0.76
Computer crime	10	Intrusion detection	0.71
Cyber threats	9	Symbolic reasoning	0.69
Network intrusion	9	Deep neural networks	0.67
Neuro-symbolic ai	9	Network intrusion	0.65
Symbolic reasoning	9	Real-time	0.65
Deep neural networks	8	Neuro-symbolic ai	0.63
Reinforcement learning	8	Artificial intelligence	0.61
Internet of things	7	Neural networks	0.59

Centrality represents degree centrality values ranging from 0 to 1

Cluster 2: Network Security and Threat Detection (n=13) captures the primary application layer, centered on network security, anomaly detection, intrusion detection systems, and threat intelligence. The presence of neuro-symbolic approaches in this cluster confirms their operational relevance in real-time security environments.

Cluster 3: AI Foundations and Interpretability (n=12) focuses on symbolic reasoning, explainability, decision making, and knowledge representation. This cluster highlights the role of transparency and human–AI interaction as defining strengths of neuro-symbolic cybersecurity systems.

Cluster 4: Complex Systems and Network Infrastructure (n=11) addresses deployment-oriented concerns such as complex networks, scalability, integration, and network intrusion, linking architectural foundations with real-world cybercrime scenarios. The identified cluster structure demonstrates an increasing convergence among learning architectures, security operations, interpretability mechanisms, and system infrastructure. This convergence defines the intellectual backbone of the field and clarifies how neuro-symbolic AI operationalizes both learning and reasoning across cybersecurity domains, thus answering RQ1 about the thematic and structural foundations.

4.5 Bibliographic coupling and intellectual structure analysis

While keyword co-occurrence reveals thematic content, bibliographic coupling exposes the intellectual structure of the field by linking publications that share common references. Two documents are bibliographically coupled if they cite the same prior work, indicating conceptual proximity and shared methodological foundations. This analysis is particularly effective for identifying influential studies, emerging schools of thought, and the evolutionary organization of a research domain.

The bibliographic coupling network is visualized in Fig. 5 where the nodes correspond to individual publications, the size of the node reflects the total strength of the link (coupling intensity), and the thickness of the edge represents the extent of shared references between studies. Color-coded clusters denote communities of intellectually related research streams. The network reveals several dominant and structurally important hubs. Studies by Piplai et al.[1, 27, 28] and Kalutharage et al.[29] appear as central anchors, indicating strong coupling with multiple works in autonomous cyber defense and knowledge-enhanced learning. Their positioning suggests that early neuro-symbolic security architectures provide a foundational knowledge base reused across later research. Similarly, the Loevenich et al.[30] lineage forms a tightly connected cluster linking reinforcement learning, knowledge graphs, and neuro-symbolic agents, reflecting the increasing role of adaptive policy reasoning in cyber operations.

Cross-cluster links demonstrate that the field is not fragmented, but rather organized around a small number of methodological cores that bridge application domains such as IoT security, threat intelligence, malware analysis, and autonomous cyber defense. The presence of inter-cluster coupling further indicates that learning–reasoning (L–R) and reasoning-for-learning (R4L) paradigms are converging across subfields instead of evolving independently.

From an evolutionary perspective, bibliographic coupling highlights the progression from early conceptual integrations toward operational neuro-symbolic systems capable of real-time reasoning, explainability, and adaptive defense. This structural coherence provides quantitative evidence that neuro-symbolic AI in cybersecurity is maturing into a unified research paradigm rather than a collection of isolated case studies, directly supporting the investigation of RQ2 concerning influential documents, research continuity, and intellectual evolution.

This combined temporal, thematic, and bibliographic coupling analysis provides a multi-layered understanding of neuro-symbolic AI in cybersecurity. Temporal trends reveal rapid maturation, keyword networks expose conceptual and application themes, and bibliographic coupling uncovers the

intellectual organization and influence pathways shaping the field. Together, these findings respond to RQ1 and RQ2 by characterizing what the field studies, how it evolves, and which works structurally define its development, while also motivating the challenge-driven discussion presented in response to RQ3 in the following section.

5 Literature review: neuro-symbolic AI applications in cybersecurity

This section presents a systematic synthesis of the state of the art in neuro-symbolic AI (NSAI) for cybersecurity, addressing RQ3 by examining existing approaches, limitations, and emerging opportunities. Based on an in-depth analysis of 63 high-quality studies, the review consolidates how neural learning and symbolic reasoning are jointly operationalized to address complex cyber threats.

The literature is organized according to three complementary perspectives: (i) architectural integration of learning and reasoning, (ii) application of NSAI across cybersecurity domains, and (iii) neural and symbolic components that enable practical deployment. This structure facilitates a holistic understanding of how NSAI transitions from conceptual integration paradigms to domain-specific real-world security solutions.

5.1 Integration architectures in neuro-symbolic cybersecurity

We begin by examining how NSAI systems integrate neural learning with symbolic reasoning. Following the taxonomy defined in Sect. 2.3, the 63 reviewed studies are grouped into three integration paradigms. Learning for Reasoning (L4R) dominates the literature with 38 studies (60.3%), followed by bidirectional Learning–Reasoning (L–R) with 18 studies (28.6%). In contrast, Reasoning for Learning (R4L) remains comparatively underexplored, accounting for only 7 studies (11.1%).

5.1.1 Learning for reasoning (L4R)—dominant paradigm

The L4R architecture represents the most prevalent approach, accounting for 60.3% of surveyed implementations. In this sequential architecture, neural components extract features or detect patterns from raw security data, while symbolic modules provide logical reasoning, rule validation, or explainable decision making. Table 10 presents representative L4R implementations across different cybersecurity domains.

Table 10 presents the most recent ten L4R applications in cybersecurity. These studies address major security domains such as IoT network protection, malware analysis, fraud

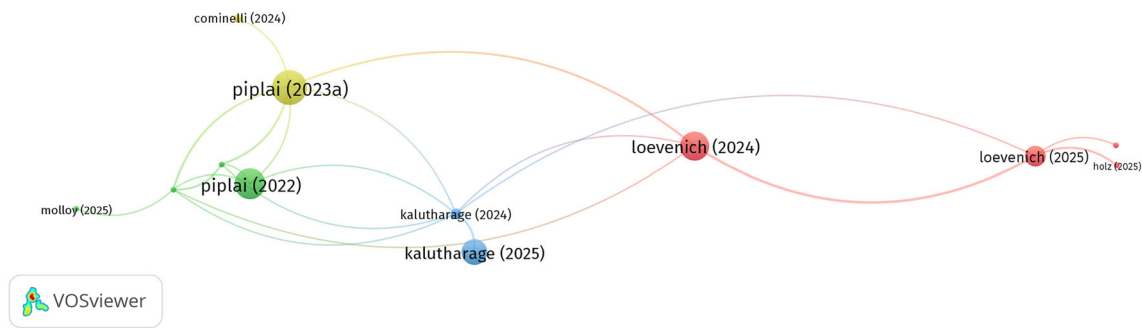


Fig. 5 Coupling network revealing intellectual structure and influential research streams in neuro-symbolic AI cybersecurity

detection, and industrial control systems. Across these works, the neural components vary according to task demands and commonly include convolutional networks, transformers, deep neural models, and autoencoder-based representations. On the symbolic side, most approaches rely on rule-based reasoning enriched with knowledge bases, ontologies, or logical formalisms to capture and exploit domain knowledge.

Recent work shows a clear shift toward addressing emerging security challenges. Privacy-preserving methods such as federated learning [31], zero-trust frameworks for IoT-based vehicles [32], and open-set recognition techniques [2] illustrate how the field is adapting to a changing threat landscape. At the same time, there is sustained attention to explainability across applications, ranging from transparent decision processes in fraud analysis to domain knowledge-guided explanation models for attack detection. This trend highlights the practical importance of interpretable security mechanisms for deployment in operational environments.

L4R architectures demonstrate significant strengths in cybersecurity applications, particularly in their ability to provide interpretable security decisions. The symbolic components within these architectures effectively translate neural network outputs into human-understandable rules and logic, addressing the critical explainability challenge in AI-driven security systems. Furthermore, L4R frameworks excel at integrating domain knowledge, allowing expert knowledge and established security policies to be systematically encoded within the symbolic reasoning layer. This integration ensures that the system's decision-making process aligns with security best practices and organizational requirements. Additionally, the transparent reasoning chain inherent in L4R architectures provides substantial support for audit requirements and compliance frameworks, making these systems particularly valuable in regulated environments where decision accountability and traceability are essential.

5.1.2 Reasoning for learning (R4L)—knowledge-guided learning

The R4L paradigm, representing 11.1% of implementations, uses symbolic knowledge to guide and constrain neural learning processes. In this architecture, symbolic components inject domain constraints, policies, and structural priors into model training, improving robustness and data efficiency. Table 11 presents representative R4L implementations across different cybersecurity domains.

Knowledge graph-augmented LLMs [44] combine reasoning capabilities with structured domain knowledge for threat detection, while multi-agent reinforcement learning with formal verification [39] ensures learned defense strategies satisfy security constraints. Knowledge-infused GANs [42] demonstrate how symbolic knowledge guides synthetic data generation, addressing limited labeled datasets. R4L architectures offer distinct advantages where domain knowledge and constraints are critical. Symbolic guidance improves data efficiency by encoding expert knowledge, reducing training data requirements in scenarios where labeled security data is scarce.

5.1.3 Learning–reasoning (L–R): bidirectional integration

The Learning–Reasoning (L–R) paradigm is characterized by iterative and bidirectional interaction between neural and symbolic components, accounting for approximately 28.6% of reported implementations. Unlike sequential pipelines, L–R architectures establish closed feedback loops in which learned representations continuously inform symbolic reasoning, while logical constraints and knowledge bases, in turn, guide neural adaptation. This reciprocal coupling enables progressive refinement across perception, reasoning, and decision layers, improving robustness under evolving threat conditions.

Table 10 Representative L4R implementations in cybersecurity

Study	Domain	Neural component	Symbolic component	Key innovation
Almadhor et al. [33]	IoT Network Security	ANN/CNN	Rule-based reasoning	Dual-model architecture for explainable intrusion detection
Islam et al. [34]	Risk Management	DNN	Knowledge-based reasoning	Vulnerability prioritization with interpretability
Ding et al. [35]	Threat Intelligence	Transformer models	Rule-based & knowledge augmentation	CTI summarization with domain knowledge
Kalutharage et al. [36]	IoT Attack Detection	Autoencoder	Rules & expert knowledge	Domain knowledge-driven explainable AI
Bizzari et al. [2]	Open Set Recognition	Autoencoder	Rule-based reasoning with XGBoost-based explainability	Novel intrusion detection with open-set capability
Molloy et al. [37]	Malware Analysis	Deep CNN	Logic-based reasoning	Zero-day malware similarity detection
Achar et al. [31]	IoT Botnet Detection	Federated deep learning	Rule-based & ontology	Privacy-preserving distributed learning
Ullah et al. [32]	Vehicle Networks	Transformer & Federated Learning	Rule-based reasoning	Zero-trust architecture for IoV
Rasheed et al. [38]	Fraud Detection	Neural Networks	Rule-based reasoning	Real-time transparent decision making

Table 11 Representative R4L implementations in cybersecurity

Study	Domain	Neural component	Symbolic component	Key innovation
Loevenich et al. [44]	Autonomous Cyber Defense	DRL Models	Knowledge Graph	KG-augmented LLM for threat detection
Holz et al. [39]	Autonomous Cyber Defense	Multi-Agent RL	KG + Formal Verification	Robust defense with verified constraints
Depa et al. [40]	Traffic Management	Transfer Learning	Rule-based System	Hybrid edge intelligence for IDS
Lei et al. [41]	Penetration Testing	Reinforcement Learning	First-Order Logic	ADAPT: Game-theoretic framework
Kotal et al. [42]	IoT Intrusion Detection	GAN	Knowledge Graph	KiNETGAN: Knowledge-infused synthetic data
Lin et al. [43]	Malware Detection	GNN	Knowledge Graph	KG-guided deep RL
Piplai et al. [28]	Malware Detection	Reinforcement Learning	Knowledge Graph	Offline RL with cybersecurity KG

In cybersecurity, L–R systems are particularly effective for autonomous and adaptive defense, where real-time situational awareness must be aligned with structured policies and expert knowledge. As summarized in Table 12, the most recent ten representative approaches integrate deep neural models with rule engines, ontologies, knowledge graphs, and formal logic to support proactive detection, forensic analysis, vulnerability assessment, and attack–defense learning. The

bidirectional nature of L–R architectures enables complex inference while preserving learning flexibility, making them a central paradigm for intelligent cyber defense systems.

Compared with L4R, L–R better supports autonomous and adaptive defense under evolving threat conditions. By balancing learning flexibility with symbolic control, bidirectional architectures represent a critical direction for building self-managing, policy-aware cyber defense infrastructures.

Table 12 Representative L–R implementations in cybersecurity

Study	Domain	Neural net	compo- nent	Symbolic compo- nent	Key innovation
Samaddar et al. [45]	Autonomous Cyber Defense	PNN		Behavior Tree	OOD detection with symbolic planning
Das et al. [46]	Consumer Electronics	LSTM		Rule-based Reasoning	Proactive security prediction
Ali et al. [47]	Internet of Energy	DNN		Knowledge-based Reasoning	Cognitive quantum resilient framework
Awan et al. [48]	Network Integration	GRU / LSTM		Rule-based Reasoning	Adaptive neuro-symbolic network
Parsonage et al. [49]	Digital Forensics	LLM		ASP + Ontology	Automated forensic analysis
Ares-Robledo et al. [50]	IoT Anomaly Detection	VGAE with GCN		Fuzzy Logic	Hyperparameter-optimized graph learning
Nalluri et al. [51]	Threat Intelligence	GNN, LSTM, HMM, RL		Rule-based Reasoning	Multi-component predictive framework
Velasco et al. [52]	Software Vulnerability	Language Models		Rule-based Reasoning	Neuro-symbolic program comprehension
Rajpal et al. [53]	Cryptography	Generative AI		Logic-based Reasoning	Quantum-resistant encryption ecosystem
Lin et al. [54]	ERP Systems	TGNN		OWL2 Ontology + SWRL	Quantum-safe ERP framework

5.2 Application domain analysis

Neuro-symbolic AI adoption in cybersecurity exhibits pronounced asymmetries across application domains. Our systematic review uncovers both strong concentration in a small set of detection-centric problems and systematic underrepresentation of several operationally critical contexts. As illustrated in Fig. 6, the majority of NSAI research targets network- and malware-centric defenses, whereas areas such as digital forensics, enterprise security, and security orchestration remain comparatively neglected.

To structure this heterogeneity, we categorize domains into three maturity tiers based on research volume and methodological depth: *mature* domains ($n \geq 6$), where NSAI techniques are repeatedly validated; *emerging* domains ($n = 3 - 5$), marked by architectural diversification and tighter learning–reasoning coupling; and *nascent* domains ($n \leq 2$), which remain largely exploratory. Within each tier, we analyze integration paradigm preferences (L4R, L–R, R4L), dominant technical objectives, and persistent research gaps that constrain operational deployment.

5.2.1 Research distribution and domain maturity

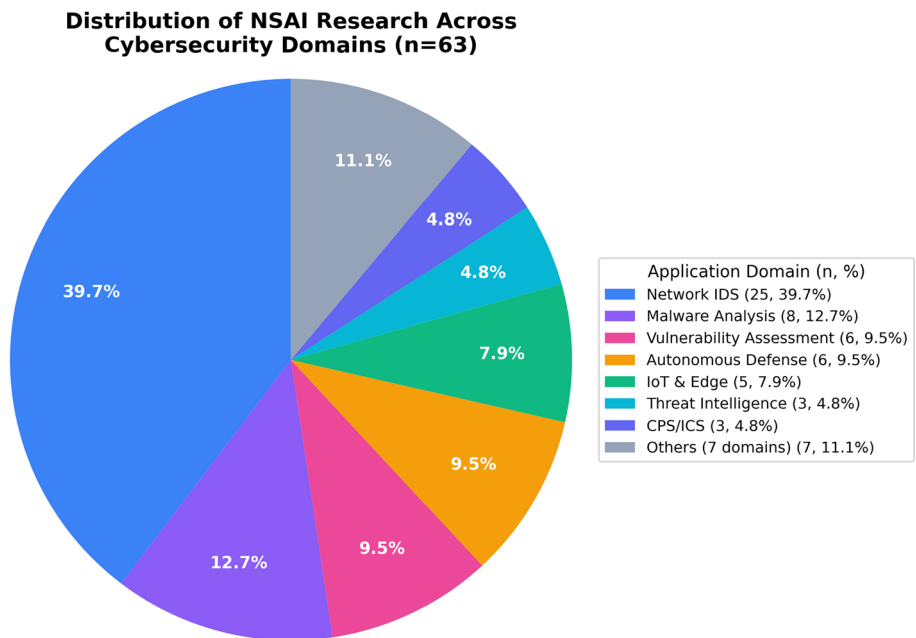
Three established domains, namely *Network Intrusion Detection* (39.7%), *Malware Analysis* (12.7%), and *Vulnerability Assessment* (9.5%), together account for 61.9% of the reviewed studies, reflecting both methodological maturity and a substantial base of empirical validation. In con-

trast, four developing domains, including *Autonomous Cyber Defense*, *IoT and Edge Security*, *Threat Intelligence*, and *Cyber-Physical Systems*, comprise 27.0% of the literature and exhibit active exploration of architectural and methodological alternatives. More critically, eight early-stage domains attract only 15.9% of research attention, despite constituting more than half of the identified application areas. This imbalance indicates systematic underrepresentation of operationally relevant contexts such as digital forensics, enterprise security, and fraud detection.

Mature domains: from feasibility to deployment

Network intrusion detection dominates the landscape ($n = 25$), corresponding to the “Network Security” and “Threat Detection & Response” subdomains in Table 3. The strong preference for L4R architectures (72%) reflects operational requirements for real-time detection combined with transparent decision making. Recent implementations demonstrate notable deployment diversity, including lightweight models for resource-constrained IoT environments [31, 33, 36], logic-enhanced transformers for safety-critical industrial control systems [55, 56], and uncertainty-aware approaches for open-set recognition of novel attacks [2]. However, the limited exploration of bidirectional L–R architectures (16%) restricts adaptability against evolving adversaries, indicating that most work still prioritizes feasibility over continuous operational refinement.

Malware Analysis ($n = 8$, 12.7%) emphasizes interpretable zero-day detection and knowledge-enhanced classification. The Mecha framework exemplifies this trend

Fig. 6 Application domains in cybersecurity

by fusing deep CNN features with logic programming for similarity-based detection [37], while knowledge graph-augmented systems encode semantic relationships among malware families and attack patterns [28, 43]. Moreover, federated learning combined with symbolic reasoning [57] addresses privacy constraints in cross-organizational malware sharing. Despite these advances, robustness against adversarial manipulation and cross-platform generalization remain insufficiently studied.

Vulnerability assessment ($n = 6$, 9.5%) aligns with the “Vulnerability & Risk Management” subdomain in Table 3 by prioritizing transparent risk ranking rather than raw detection. Key innovations include organizationally contextualized knowledge-based ranking [34], LLM–formal analysis fusion for code-level vulnerability identification [58], and probabilistic logic programming for uncertainty-aware risk estimation [59]. While effective for compliance-driven environments, the domain still lacks automated patch generation and supply-chain–level reasoning.

Emerging domains: architectural diversity and balanced integration

Emerging application areas exhibit greater architectural heterogeneity and a more balanced coupling between learning and reasoning components. In particular, **Autonomous Cyber Defense** ($n = 6$, 9.5%) demonstrates the widest distribution of integration paradigms, with L–R (50%), R4L (33.3%), and L4R (16.7%). This dispersion suggests that effective adaptive response mechanisms benefit from tighter bidirectional interaction between neural and symbolic layers. Representative approaches integrate behavior trees for out-of-distribution detection [45], knowledge graph–enhanced deep reinforcement learning agents [30], and multi-agent

architectures augmented with formal verification to ensure the safety and reliability of learned strategies [39]. Collectively, these designs indicate that response-centric cybersecurity applications require richer learning–reasoning feedback loops than those typically employed in detection-focused tasks.

IoT & Edge Security ($n = 5$, 7.9%) remains comparatively underdeveloped given the rapid proliferation of connected devices. Existing work explores zero-trust vehicular networks [32], federated botnet detection under privacy constraints [31], and fuzzy-logic graph learning for resource-limited platforms [50]. However, most approaches adapt heavyweight NSAI pipelines rather than designing hardware-aware, lightweight reasoning architectures suitable for edge environments.

Threat Intelligence and Cyber-Physical Systems (each $n = 3$, 4.8%) remain at early stages of exploration. Threat intelligence research centers on CTI summarization and predictive modeling [35, 51] but lacks real-time stream integration and geopolitical context reasoning. CPS-oriented work emphasizes industrial control protection and formal verification [55, 56], yet the small volume of studies is disproportionate to the risks associated with critical infrastructure, motivating the need for certified and verifiable NSAI systems.

Under-explored domains

The long tail of nascent domains ($n \leq 2$) represents substantial untapped potential for NSAI. **Digital Forensics** is particularly well suited to symbolic reasoning over evidence chains and temporal reconstruction, yet only isolated efforts integrate LLMs with ontology-driven analysis [49]. **Penetration Testing** uniquely applies reinforcement learning guided

by logical constraints [41], suggesting broader opportunity for constraint-aware exploration strategies.

Similarly, **Fraud Detection** and **Enterprise Security** remain sparsely investigated despite regulatory demands for explainability and policy compliance [38, 54]. **Social Media Security** lacks principled reasoning over social graphs for misinformation and influence operations, while **Cryptography & Privacy** and **Phishing Detection** show limited progress toward privacy-preserving and multimodal NSAI pipelines. The skewed distribution in Fig. 6 thus reflects not only historical research focus but also significant unrealized potential for NSAI in cybersecurity practice.

5.3 Challenges and future research directions

Despite growing interest and promising results in neuro-symbolic approaches for cybersecurity, the field remains at an early stage of operational maturity. Most existing studies are evaluated under controlled experimental settings and focus primarily on detection accuracy rather than broader requirements of real-world deployment. This subsection synthesizes principal challenges and articulates concrete future research directions for advancing NSAI-based cybersecurity systems toward robust, deployable solutions.

5.3.1 Computational efficiency and scalability

Real-time cybersecurity requires processing massive volumes of heterogeneous data with strict latency constraints. Deep architectures impose substantial computational overhead in execution time, memory consumption, and hardware requirements. Neuro-symbolic systems can address this by using symbolic reasoning to pre-filter, structure, or prioritize data before neural processing, encoding domain knowledge such as protocol constraints and attack patterns. However, symbolic reasoning mechanisms suffer from scalability limitations with large, evolving knowledge graphs, and coordinating symbolic-neural components introduces design and runtime overhead.

A critical limitation in reviewed literature is the lack of detailed reporting on computational costs. While many works focus on accuracy and false-positive rates, they rarely provide comprehensive information on execution time, memory usage, hardware configurations, or performance under realistic traffic conditions. Future research must develop lightweight hybrid architectures optimized for edge deployment, establish standardized efficiency benchmarks, and achieve optimal balance between symbolic and neural components.

5.3.2 Explainable artificial intelligence (XAI) in cybersecurity

Transparency is essential for trust, auditing, and operational control in cybersecurity. While neuro-symbolic architectures promise better interpretability than purely data-driven deep learning models, producing reliable and useful explanations for real deployments remains an open problem. Interpretable approaches [60, 61] are commonly associated with limited scalability and, in many cases, inferior detection performance compared to deep learning models. Their representational capacity is restricted when dealing with highly non-linear and heterogeneous cyber data. By contrast, black-box architectures achieve stronger predictive accuracy but lack transparency, necessitating post-hoc explanation mechanisms.

Existing post hoc methods can be grouped into: feature-oriented techniques estimating input attribute importance [62, 63], decomposition-based methods tracing predictions through internal structures [64, 65], and perturbation-based explanations [66]. In reviewed literature, post hoc explanation techniques such as SHAP and LIME frequently complement neural predictions [33, 34, 67]. Several works employ rule-based explanations to expose symbolic inference paths [54, 68], while [48] enriches explanations with contextual information through symbolic reasoning engines.

An important research challenge is harmonizing intrinsic reasoning provided by neuro-symbolic architectures with explanation mechanisms that translate technical inferences into actionable security insight. Future work should develop unified explainability frameworks integrating symbolic reasoning throughout the decision pipeline, design multi-level explanations tailored to diverse stakeholders, and establish evaluation metrics assessing explanation utility beyond technical accuracy.

5.3.3 Limited availability of high-quality datasets

Limited availability of high-quality annotated datasets constrains cybersecurity research. The reviewed literature exhibited heavy concentration on limited benchmark datasets, with notable reliance on CIC-IDS2017 and UNSW-NB15, appearing in five and four studies respectively. While most datasets were publicly available, a substantial portion remained inaccessible through self-curated or proprietary industrial data, constraining independent validation and cross-study comparison.

Table 13 presents a comprehensive inventory of datasets employed across reviewed studies, classified by origin and attack taxonomy. The analysis reveals three distinct categories: publicly accessible benchmark datasets (CIC-IDS2017, UNSW-NB15, NF-BoT-IoT-V2, BoTNeT-IoT-L01, CVEjoin, Devign, CWE-Bench-Java, MalwareBazaar, Dike-

Dataset, VirusShare), self-curated datasets for specific research objectives (vulnerability data from National Vulnerability Database [59], synthetic attack-defense scenarios [27], threat intelligence corpora [35]), and proprietary datasets from operational industrial environments [54] providing high ecological validity but constraining independent validation.

Beyond availability concerns, dataset quality documentation presents significant challenges. Most studies failed to report essential characteristics such as dataset size, class balance, preprocessing procedures, or data leakage control measures. This lack of methodological transparency makes it difficult to assess whether observed performance improvements stem from genuine algorithmic advances or from artifacts of data preparation and experimental setup. The absence of preprocessing details is particularly problematic for neuro-symbolic approaches, which require careful integration of symbolic knowledge with raw data.

Addressing dataset challenges requires coordinated community effort. Critical needs include establishing standardized benchmark suites specifically designed for neuro-symbolic evaluation with diverse attack taxonomies and explicit symbolic knowledge bases, adopting rigorous dataset documentation standards, encouraging systematic dataset reuse through cross-validation on multiple benchmarks, and developing synthetic data generation frameworks producing realistic, privacy-preserving security datasets with controlled characteristics.

5.3.4 Future research frontiers

Advancing explainability for operational trust. Future research should develop unified explainability frameworks that seamlessly integrate symbolic reasoning with neural learning throughout the decision pipeline, rather than treating explanation as post-hoc addition. This includes designing explanation-aware architectures where symbolic components actively generate traceable reasoning chains, investigating methods for producing multi-level explanations tailored to different stakeholder needs, and establishing evaluation metrics assessing explanation quality including operational utility, cognitive load, and decision-support effectiveness.

Strengthening adversarial robustness. Cybersecurity environments are characterized by adaptive adversaries who continuously evolve attack strategies, yet reviewed literature provides limited investigation of neuro-symbolic systems' robustness against adversarial manipulation. Future research must prioritize developing systems with inherent adversarial robustness, investigating how symbolic reasoning can detect and mitigate adversarial perturbations, exploring online learning mechanisms enabling continuous adaptation, and examining how formal verification techniques ensure learned policies maintain safety properties when adapted to novel threats.

Causal reasoning and counterfactual analysis. Traditional cybersecurity analytics rely on correlation-based pattern recognition without establishing directional causality. A critical research frontier involves integrating causal reasoning capabilities into neuro-symbolic architectures to enable deeper understanding of attack causality, support counterfactual scenario analysis, and facilitate instructable refinement aligned with organizational objectives. Future research should prioritize developing automated causal discovery algorithms tailored for cybersecurity domains, real-time causal inference frameworks, formal verification methods for causal explanations, and interactive interfaces enabling security analysts to validate and refine causal models.

Quantum-enhanced architectures. Emerging research at the intersection of quantum computing, NSAI, and cybersecurity presents transformative opportunities to tackle computationally intractable security challenges. Klusch et al. [69] highlight how quantum-based optimization and reasoning techniques can serve as foundational components of hybrid neuro-symbolic architectures. Future research should focus on hybrid quantum-classical NSAI architectures leveraging quantum resources for computational bottlenecks while maintaining interpretability through symbolic reasoning, quantum-enhanced knowledge graph reasoning for large-scale threat intelligence, and practical deployment strategies addressing hardware limitations and error correction.

Ethical, legal, and societal implications. Beyond technical challenges, NSAI deployment in cybersecurity raises critical ethical, legal, and societal concerns. Privacy preservation remains paramount, particularly when symbolic reasoning processes sensitive organizational data or when federated learning aggregates threat intelligence across jurisdictions with divergent data protection regimes. Adversarial robustness carries ethical dimensions: adversarial manipulation of symbolic knowledge bases could systematically bias threat assessment [70]. Legal compliance frameworks must address accountability gaps in hybrid neural-symbolic decision making. Furthermore, asymmetric accessibility of advanced NSAI capabilities risks widening security gaps between well-resourced organizations and under-resourced critical infrastructure operators, necessitating policy frameworks promoting equitable access while preventing malicious exploitation.

5.4 Limitations

While this study provides comprehensive scientometric mapping of NSAI in cybersecurity, certain methodological trade-offs merit acknowledgment. The single-database approach using Scopus was deliberately chosen to ensure metadata consistency, citation accuracy, and analytical reproducibility. However, this prioritizes quality over breadth and may underrepresent gray literature, preprint repositories, and some regional conferences. The final corpus ($n = 63$) reflects

Table 13 Dataset inventory for neuro-symbolic cybersecurity

Dataset name	Origin	Attack types	Used in studies
CIC-IDS2017	Real	DoS, DDoS, Botnet, Web Attacks	[32, 36, 38, 47, 50]
UNSW-NB15	Hybrid	Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms	[36, 49–51]
NF-BoT-IoT-V2	Real	DDoS, DoS, Reconnaissance, Theft	[33]
BoTNeT-IoT-L01	Real	Botnet, DDoS	[31]
CICIOTDIAD2024	Real	DDoS, DoS, Reconnaissance, MQTT attacks, Spoofing	[32]
CICIOV2024	Real	DoS, Spoofing	[32]
IoTID20	Real	DoS, MITM	[50]
USBIDS	Synthetic	DoS attack patterns	[36]
CVEjoin	Real	Vulnerabilities, threat behavior	[34]
CWE-Bench-Java	Real	Path Traversal, OS Command Injection, XSS, Code Injection	[58]
Devign	Real	C++ vulnerabilities	[52]
NVD-derived	Self-curated	Vulnerability risk data	[59]
CTISum	Real	Threat intelligence content	[35]
CTI-Classification	Real	Attack techniques	[44]
MalwareBazaar, DikeDataset, VirusShare	Real	Ransomware, Trojan, Botnet, Worms, Downloaders, Spyware	[43]
ERP Industrial	Proprietary	Manufacturing, supply chain, schema matching	[54]
Attack-Defense	Self-curated	Simulated attack scenarios	[27]

rigorous quality filtering to ensure genuine neuro-symbolic integration rather than superficial "hybrid AI" claims. This selective approach limits statistical power for fine-grained subgroup analyses and may not capture unpublished industrial implementations. Future research could address these limitations through multi-database validation and broader corpus construction to enhance generalizability.

6 Conclusion

This study provides the first comprehensive scientometric mapping of neuro-symbolic AI research in cybersecurity, analyzing 262 publications (2016–2025) with systematic review of 63 high-quality studies. Our findings demonstrate that NSAI represents a fundamental advancement through principled integration of neural adaptability with symbolic reasoning, addressing the critical need for both performance and interpretability in cybersecurity systems.

The field exhibits exponential growth (162% increase in 2024–2025) and organized intellectual structure, with four thematic clusters showing strong interconnectivity (modularity $Q = 0.170$). Learning-for-Reasoning architectures dominate (60.3%), though bidirectional Learning–Reasoning frameworks (28.6%) show promise for adaptive cyber defense. Network intrusion detection and malware analysis emerge as mature domains, while autonomous

defense, IoT security, and digital forensics remain under-explored despite operational significance.

Our key contributions include: (1) rigorous bibliometric analysis revealing temporal trends and intellectual structures; (2) systematic taxonomies of integration architectures, neural components, and symbolic frameworks; (3) identification of critical gaps including computational efficiency benchmarking, standardized evaluation frameworks, and explainability integration. To support reproducibility, we provide complete bibliographic data, network analysis artifacts, and methodological templates through a dedicated repository.

Advancing NSAI-driven cybersecurity requires three coordinated efforts: establishing evaluation benchmarks assessing both detection performance and computational efficiency; developing unified explainability frameworks integrating symbolic reasoning throughout decision pipelines; and expanding research to underserved domains where interpretability directly addresses regulatory compliance needs. The substantial promise of NSAI will be realized through sustained commitment to building hybrid intelligence frameworks that strengthen cybersecurity infrastructure while maintaining the transparency and trustworthiness essential for operational acceptance. All data and resources supporting this study are publicly available at: <https://github.com/ShamaSharma/Charting-the-evolution-of-Neuro-symbolic-AI-in-cybersecurity>

Author Contributions S. and S.J. jointly conceptualized the study and designed the research framework. S. performed data collection, scientometric analysis, and prepared the initial draft of the manuscript. S.J. provided supervision and guided the manuscript revisions. Both authors reviewed and approved the final version of the manuscript.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

- Piplai, A., Kotal, A., Mohseni, S., Gaur, M., Mittal, S., Joshi, A.: Knowledge-enhanced neurosymbolic artificial intelligence for cybersecurity and privacy. *IEEE Internet Comput.* **27**(5), 43–48 (2023)
- Bizzarri, A., Yu, C., Jalaian, B., Riguzzi, F., Bastian, N.: Neuro-symbolic integration for open set recognition in network intrusion detection. In: *International Conference of the Italian Association for Artificial Intelligence*, pp. 50–63 (2024)
- De Raedt, L., Dumančić, S., Manhaeve, R., Marra, G.: From statistical relational to neuro-symbolic artificial intelligence. *arXiv preprint arXiv:2003.08316* (2020)
- Hitzler, P., Eberhart, A., Ebrahimi, M., Sarker, M.K., Zhou, L.: Neuro-symbolic approaches in artificial intelligence. *Natl. Sci. Rev.* **9**(6), 035 (2022)
- Wang, W., Yang, Y., Wu, F.: Towards data-and knowledge-driven ai: a survey on neuro-symbolic computing. *IEEE Trans. Pattern Anal. Mach. Intell.* (2024)
- Garcez, A.D., Lamb, L.C.: Neurosymbolic ai: the 3rd wave. *Artif. Intell. Rev.* **56**(11), 12387–12406 (2023)
- Sheth, A., Roy, K., Gaur, M.: Neurosymbolic artificial intelligence (why, what, and how). *IEEE Intell. Syst.* **38**(3), 56–62 (2023)
- Jalaian, B., Bastian, N.: Neurosymbolic ai in cybersecurity: bridging pattern recognition and symbolic reasoning. In: *MILCOM 2023-IEEE Military Communications Conference*, pp. 268–273 (2023)
- Yu, D., Yang, B., Liu, D., Wang, H., Pan, S.: A survey on neural-symbolic learning systems. *Neural Netw.* **166**, 105–126 (2023)
- Zhang, X., Sheng, V.S.: Bridging the gap: representation spaces in neuro-symbolic ai. *arXiv preprint arXiv:2411.04393* (2024)
- Hagos, D.H., Rawat, D.B.: Neuro-symbolic ai for military applications. *IEEE Trans. Artif. Intell.* (2024)
- Chen, W., Ma, X., Wang, Z., Li, W., Fan, C., Zhang, J., Que, X., Li, C.: Exploring neuro-symbolic ai applications in geoscience: implications and future directions for mineral prediction. *Earth Sci. Inf.* **17**(3), 1819–1835 (2024)
- Hamilton, K., Nayak, A., Božić, B., Longo, L.: Is neuro-symbolic ai meeting its promises in natural language processing? A structured review. *Semantic Web* **15**(4), 1265–1306 (2024)
- Wan, Z., Liu, C.-K., Yang, H., Li, C., You, H., Fu, Y., Wan, C., Krishna, T., Lin, Y., Raychowdhury, A.: Towards cognitive ai systems: a survey and prospective on neuro-symbolic ai. *arXiv preprint arXiv:2401.01040* (2024)
- Bhuyan, B.P., Ramdane-Cherif, A., Tomar, R., Singh, T.: Neuro-symbolic artificial intelligence: a survey. *Neural Comput. Appl.* **36**(21), 12809–12844 (2024)
- Hakim, S.B., Adil, M., Velasquez, A., Xu, S., Song, H.: Neuro-symbolic ai for cybersecurity: state of the art, challenges, and opportunities. *arXiv preprint arXiv:2509.06921* (2025)
- De Raedt, L., Kersting, K.: Probabilistic logic learning. *ACM SIGKDD Explor. Newsl.* **5**(1), 31–48 (2003)
- Kautz, H.: The third ai summer: Aai 2020 presidential address. *AI Mag.* **41**(3), 3–5 (2020)
- Besold, T.R., Garcez, A.d., Bader, S., Bowman, H., Domingos, P., Hitzler, P., Kühnberger, K.-U., Lamb, L.C., Lowd, D., Lima, P., et al.: Neural-symbolic learning and reasoning: a survey and interpretation. *J. Artif. Intell. Res.* **51**, 1–87 (2017)
- Marcus, G., Davis, E.: *Rebooting AI: Building Artificial Intelligence We Can Trust*. Pantheon Books (2019)
- Rahmati, M.: Towards explainable and lightweight ai for real-time cyber threat hunting in edge networks. *arXiv preprint arXiv:2504.16118* (2025). Preprint, submitted April 18, 2025
- Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., et al.: The prisma 2020 statement: an updated guideline for reporting systematic reviews. *bmj* **372** (2021)
- Martinez, P., Al-Hussein, M., Ahmad, R.: A scientometric analysis and critical review of computer vision applications for construction. *Autom. Constr.* **107**, 102947 (2019)
- Haghani, M.: What makes an informative and publication-worthy scientometric analysis of literature: a guide for authors, reviewers and editors. *Transport. Res. Interdiscip. Perspect.* **22**, 100956 (2023)
- Van Eck, N., Waltman, L.: Software survey: Vosviewer, a computer program for bibliometric mapping. *Scientometrics* **84**(2), 523–538 (2010)
- Waltman, L., Van Eck, N.J.: A smart local moving algorithm for large-scale modularity-based community detection. *Eur. Phys. J. B* **86**(11), 471 (2013)
- Piplai, A., Anoruo, M., Fasaye, K., Joshi, A., Finin, T., Ridley, A.: Knowledge guided two-player reinforcement learning for cyber attacks and defenses. In: *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)*, pp. 1342–1349 (2022). IEEE
- Piplai, A., Joshi, A., Finin, T.: Offline rl+ ckg: a hybrid ai model for cybersecurity tasks. In: *AAAI 2023 Spring Symposium on ML and Knowledge Engineering* (2023)
- Kalutharage, C., Liu, X., Chrysoulas, C., Bamgboye, O.: Neurosymbolic learning in the xai framework for enhanced cyberattack detection with expert knowledge integration. In: *IFIP International Conference on ICT Systems Security and Privacy Protection*, pp. 236–249 (2024)
- pared, J., Adler, E., Huerten, T., Lopes, R.: Design and evaluation of an autonomous cyber defence agent using drl and an augmented llm. *Comput. Netw.* **262**, 111162 (2025)
- Achar, H.B., Prasad, P., Rajeshwari, D., Raghu, N., Hemanth Kumar, B., Mahesh, T.: Federated learning and neuro-symbolic ai for botnet classification in iot-enabled consumer devices. *Int. J. Inf. Technol.*, pp. 1–8 (2025)
- Ullah, F., Srivastava, G., Mostarda, L., Raza, U.: Ztid-iov: Zero-trust intrusion detection in iov using neurosymbolic ai approach with federated meta-learning. *IEEE Trans. Consumer Electron.* (2025)
- Almadhor, A., Alsubai, S., Hejaili, A.A., Klai, Z., Bouallegue, B., Kovac, U.: Designing a neuro-symbolic dual-model architecture for explainable and resilient intrusion detection in iot networks. *Sci. Rep.* **15**(1), 42786 (2025)
- Islam, S., Basheer, N., Papastergiou, S., Ciampi, M., Silvestri, S.: Intelligent dynamic cybersecurity risk management framework with explainability and interpretability of ai models for enhancing security and resilience of digital infrastructure. *J. Reliab. Intell. Environ.* **11**(3), 12 (2025)
- Ding, J., Lu, Y.: Knowledge-guided cyber threat intelligence summarization via term-oriented input construction. *Electronics* **14**(15), 3096 (2025)

36. Kalutharage, C., Liu, X., Chrysoulas, C.: Neurosymbolic learning and domain knowledge-driven explainable ai for enhanced iot network attack detection and response. *Comput. Secur.* **151**, 104318 (2025)
37. Molloy, C., Banks, J., Ding, S.H., Alaca, F., Charland, P., Walenstein, A.: Mecha: A neural-symbolic open-set homogeneous decision fusion approach for zero-day malware similarity detection. *IEEE Trans. Softw. Eng.* (2025)
38. Rasheed, A.: Neuro-symbolic self-explanatory ai (ns-xai): a real-time framework for transparent and adaptive decision-making. *Int. J. Eng. Dev. Res.* **13**(4), 948–951 (2025)
39. Holz, L., Loevenich, J., Lopes, R.R.F.: Towards robust autonomous cyber defence agents using hybrid ai models. In: 2025 IEEE 11th International Conference on Network Softwarization (NetSoft), pp. 269–272 (2025). IEEE
40. Depa, R.R., Zhang, Y., Xu, D.: Hybrid edge intelligence for real-time intrusion detection in advanced traffic management systems. In: 2025 IEEE Security and Privacy Workshops (SPW), pp. 352–354 (2025). IEEE
41. Lei, H., Ge, Y., Zhu, Q.: Adapt: A game-theoretic and neuro-symbolic framework for automated distributed adaptive penetration testing. In: MILCOM 2024-IEEE Military Communications Conference, pp. 7–12 (2024)
42. Kotal, A., Luton, B., Joshi, A.: Kinetgan: Enabling distributed network intrusion detection through knowledge-infused synthetic data generation. In: 2024 IEEE 44th International Conference on Distributed Computing Systems Workshops (ICDCSW), pp. 140–145 (2024). IEEE
43. Lin, H.-C., Liao, Y.-W., Wang, P., Lin, W.-H., Lin, Y.-H.: Using knowledge graph and deep reinforcement learning for malware detection. In: 2024 IEEE 6th Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS), pp. 493–495 (2024). IEEE
44. Loevenich, J.F., Adler, E., Mercier, R., Velazquez, A., Lopes, R.R.F.: Design of an autonomous cyber defence agent using hybrid ai models. In: 2024 International Conference on Military Communication and Information Systems (ICMCIS), pp. 1–10 (2024). IEEE
45. Samaddar, A., Potteiger, N., Koutsoukos, X.: Out-of-distribution detection for neurosymbolic autonomous cyber agents. In: 2025 IEEE 4th International Conference on AI in Cybersecurity (ICAIC), pp. 1–9 (2025)
46. Das, D.: Synaps: A neuro-symbolic framework for proactive security in consumer electronics. *IEEE Trans. Consumer Electron.* (2025)
47. Ali, S., Nawar, I., Lee, I.H., Sajjad, M., Alotaibi, A., Albuquerque, V.H.C., Muhammad, K.: Neuroshield-ioe: a cognitive, quantum-resilient, and self-evolving cybersecurity framework for the internet of energy. *IEEE Netw.* (2025)
48. Awan, K.A., Cengiz, K., Khan, S., Alqahtani, A.M., Ivković, N., Rehman, A.: Ansin: adaptive neural symbolic network for secure terrestrial and non-terrestrial integration. In: IEEE INFOCOM 2025-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), pp. 1–6 (2025). IEEE
49. Parsonage, G., Dempsey, J.: Automated generative ai-driven forensic analysis (agafa). an explainable neuro-symbolic approach to digital forensics. In: 2025 International Conference on Software, Knowledge, Information Management and Applications (SKIMA), pp. 1–6 (2025). IEEE
50. Ares-Robledo, F., Rifa-Pous, H., Clarisó, R.: Hyperparameter optimization in neuro-symbolic unsupervised graph learning. In: International Conference on Availability, Reliability and Security, pp. 275–292 (2025). Springer
51. Nalluri, S., Malyala, M.M., Kandagiri, H., Kandagiri, K.K.: Nscit: a hybrid neuro-symbolic framework for ai-driven predictive cyber threat intelligence. In: 2025 4th International Conference on Computational Modelling, Simulation and Optimization (ICCMO), pp. 14–21 (2025). IEEE
52. Velasco, A., Garryyeva, A., Palacio, D.N., Mastropaolo, A., Poshvanyk, D.: Toward neurosymbolic program comprehension. *arXiv preprint arXiv:2502.01806* (2025)
53. Rajpal, T.S., Naithani, A.: Neurocrypt: a neuro symbolic ai ecosystem for advanced cryptographic data security and transmission. *IEEE Trans. Artif. Intell.* (2025)
54. Lin, Y., Lin, Y.: Nsea: A resilient erp framework integrating quantum-safe cryptography and neuro-symbolic reasoning for industrial adaptability. *IEEE Access* (2025)
55. Zhang, Y., Du, Y., Sun, L., Zhang, C., Lai, Y.: Dynaflo logic transformer: a neuro-symbolic approach to industrial control system intrusion detection. In: International Conference on Intelligent Computing, pp. 63–74 (2025). Springer
56. Xu, X., Lai, Y., Zhang, X.: Explaining intrusion detection in industrial control systems through rule set learning. *IEEE Netw. Lett.* **7**(3), 234–238 (2025)
57. Shalaginov, A., Franke, K.: A deep neuro-fuzzy method for multi-label malware classification and fuzzy rules extraction. In: 2017 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 1–8 (2017). IEEE
58. Li, Z., Dutta, S., Naik, M.: Iris: Llm-assisted static analysis for detecting security vulnerabilities. *arXiv preprint arXiv:2405.17238* (2024)
59. Zeng, Z., Yang, Z., et al.: Licality-likelihood and criticality: vulnerability risk prioritization through logical reasoning and deep learning. *IEEE Trans. Netw. Serv. Manage.* **19**, 1746–1760 (2021)
60. Mahbooba, B., Timilsina, M., Sahal, R., Serrano, M.: Explainable artificial intelligence (xai) to enhance trust management in intrusion detection systems using decision tree model. *Complexity* **2021**(1), 6634811 (2021)
61. Dias, T., Oliveira, N., Sousa, N., Praça, I., Sousa, O.: A hybrid approach for an interpretable and explainable intrusion detection system. In: International Conference on Intelligent Systems Design and Applications, pp. 1035–1045 (2021). Springer
62. Wang, M., Zheng, K., Yang, Y., Wang, X.: An explainable machine learning framework for intrusion detection systems. *IEEE Access* **8**, 73127–73141 (2020)
63. Islam, S.R., Eberle, W., Ghafoor, S.K., Siraj, A., Rogers, M.: Domain knowledge aided explainable artificial intelligence for intrusion detection and response. *arXiv preprint arXiv:1911.09853* (2019)
64. Amarasinghe, K., Kenney, K., Manic, M.: Toward explainable deep neural network based anomaly detection. In: 2018 11th International Conference on Human System Interaction (HSI), pp. 311–317 (2018). IEEE
65. Caforio, F.P., Andresini, G., Vessio, G., Appice, A., Malerba, D.: Leveraging grad-cam to improve the accuracy of network intrusion detection systems. In: International Conference on Discovery Science, pp. 385–400 (2021). Springer
66. Wu, C., Qian, A., Dong, X., Zhang, Y.: Feature-oriented design of visual analytics system for interpretable deep learning based intrusion detection. In: 2020 International Symposium on Theoretical Aspects of Software Engineering (TASE), pp. 73–80 (2020). IEEE
67. Gaspar, D., Silva, P., Silva, C.: Explainable ai for intrusion detection systems: lime and shap applicability on multi-layer perceptron. *IEEE Access* **12**, 30164–30175 (2024)

68. Paredes, J., Simari, G., Martinez, M., Falappa, M.: Detecting malicious behavior in social platforms via hybrid knowledge-and data-driven systems. *Futur. Gener. Comput. Syst.* **125**, 232–246 (2021)
69. Klusch, M., Lässig, J., Müssig, D., Macaluso, A., Wilhelm, F.K.: Quantum artificial intelligence: a brief survey. *KI-Künstliche Intelligenz* **38**(4), 257–276 (2024)
70. Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., Galstyan, A.: A survey on bias and fairness in machine learning. *ACM Comput. Surv.* **54**(6), 1–35 (2021)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.